Calibrated Dependence: A Framework for Student Autonomy in the Age of Generative AI

Overview

University students today are using generative AI for nearly everything, from writing essays and solving problems to consuming an endless stream of AI-powered social media content. This is just the beginning; in five years, this technology will be even more deeply woven into their lives. This presents a critical trade-off. On one hand, AI offers incredible benefits, boosting productivity and unlocking new creative possibilities. On the other hand, over-relying on these tools risks eroding essential skills like critical thinking and personal autonomy. The challenge isn't to stop students from using AI, but to help them find the right balance—to embrace the benefits without the drawbacks. This project directly addresses this challenge by introducing and testing the *Calibrated Dependence* (CD) framework. Our goal is to provide the first scientific method for identifying the "sweet spot" of AI use—a state of productive, healthy reliance—and to develop interventions that help students find and maintain it. The research has in two main phases: first, a large-scale observational study using privacy-preserving browser extensions and data donation will map the current landscape of students' real-world AI use. Second, a series of controlled experiments using a custom-built web portal will test the CD framework and the efficacy of theory-driven interventions (e.g., "micro-pause" reflection prompts) in two high-stakes contexts: academic problem-solving and social media consumption for identity formation. The expected outcomes are threefold: a validated theoretical framework and metric (the CD-Score) for analyzing the trade-offs of AI use; a set of evidence-based, domain-general design principles for creating autonomy-preserving AI scaffolds; and an open-source toolkit for researchers and instructional designers.

Intellectual Merit

The intellectual merit of this project is threefold, delivering significant theoretical, empirical, and methodological innovations. Theoretically, we introduce the Calibrated Dependence (CD) framework, which bridges a critical gap by synthesizing Self-Regulated Learning (SRL) theory with HCI principles of trust calibration to create the first testable model for AI-supported self-regulation. Empirically, we will produce the first large-scale, longitudinal dataset of real-world AI usage, moving beyond unreliable self-reports to provide a foundational resource for the research community. Methodologically, we introduce and validate the Calibrated Dependence Score (CD-Score)—a novel, domain-general metric integrating psychological and behavioral data—and deliver a set of evidence-based design patterns for autonomy-preserving AI systems, packaged as an open-source toolkit.

Broader Impacts

This project will deliver significant broader impacts by translating its research findings into tangible artifacts for three key communities. For the scientific community, we will contribute a foundational public good: the first large-scale, longitudinal dataset of real-world AI interaction and the validated CD-Score metric. These resources will enable other researchers in HCI, the learning sciences, and computational social science to move beyond unreliable self-reports and conduct new, evidence-based studies on human-AI interaction. For educators and students, we will produce the Open Calibrated Dependence Toolkit, an open-source curriculum with ready-to-use modules that directly translate our findings into classroom practice, empowering students to analyze and manage their own AI use. Finally, for the technology industry, this project offers concrete solutions, providing an open-source SDK with validated design patterns that engineers can directly implement to build more autonomy-preserving systems. By creating these resources, this work provides a direct pathway from foundational research to practical tools that can shape a more user-aligned AI future.

1 Introduction

Generative artificial intelligence has become a fundamental component of university life, fundamentally reshaping both the intellectual work and social identities of students. On one front, a majority now actively use Large Language Models (LLMs) like ChatGPT for academic production, to brainstorm essays, debug code, and summarize readings [1, 2, 3]. On another, they passively consume a torrent of AI-curated and synthetically generated content on social media platforms, from algorithmically optimized video feeds to AI-powered filters that alter their appearance [4]. If this is the landscape today, the depth of AI integration in five years will be orders of magnitude greater. This dual-front transformation presents a fundamental trade-off: the promise of unprecedented productivity and creative power versus the risk of eroding the very cognitive and emotional skills that higher education aims to develop. The core problem is that we lack a principled, quantitative framework for navigating this trade-off. Without one, educators, students, and platform designers are left at an impasse, caught between blanket bans and uncritical adoption, unable to distinguish beneficial assistance from detrimental over-dependence.

This CAREER proposal directly addresses this impasse by introducing and empirically validating the Calibrated Dependence (CD) framework. We define calibrated dependence as the dynamic equilibrium where external AI assistance maximizes functional and creative benefits while preserving a user's autonomy and well-being. The framework's central innovation is the CD-Score, a novel metric that quantifies this balance by subtracting the Autonomy Penalty (the cost to a user's agency, critical thinking, and self-perception) from the Utility Gain (the performance and affective benefits). We theorize that the relationship between the level of AI assistance and the CD-Score follows an inverted-U curve. The scientific goal of this project is to map this curve, identify its "optimal band," and test interventions that help users stay within it.

To organize the vast design space of AI interactions, the CD framework uses two universal axes: Instrumentality, which captures how mission-critical or goal-oriented a task is, and Identity Salience, which captures how strongly the activity is tied to a user's self-concept. As shown in Figure 1, mapping digital activities onto this space provides a powerful co-ordinate system for anticipating where the challenges of calibration will be most acute. For example, low-stakes activities like scrolling through memes fall in the low-instrumentality/low-identity quadrant, while high-stakes tasks like drafting a college application essay occupy the high-instrumentality/high-identity quadrant.

To ground this general framework in concrete, high-stakes contexts, this project focuses on two ubiquitous and developmentally critical use cases that were deliberately selected because they occupy distinct, complementary positions on the Calibrated Dependence Map: (i) Academic Problem-Solving: A high-instrumentality, moderate-identity context where students use AI for coursework, and the primary risk is the erosion of critical thinking and deep learning; (ii) Social Media & Body Image: A high-identity, moderate-instrumentality context where students are exposed to AI-generated images and avatars, and the primary risk is the negative psychological impact of social comparison.

This research program integrates large-scale observational methods with controlled experimentation to tackle this problem from a social computing perspective. It asks four linked questions:

RQ1 (Prevalence): How are university students truly using and being exposed to generative AI in their daily academic and social lives? To answer this, we will deploy a privacy-preserving instrumentation layer to build the first large-scale, observational dataset of real-world AI interaction, moving beyond the limitations of self-report surveys.

RQ2 (Mapping the Curve - Learning): In the context of academic work, what is the precise relationship between the level of AI assistance and the Calibrated Dependence Score? A controlled experiment in our LearnLab environment will map this curve and test how it is moderated

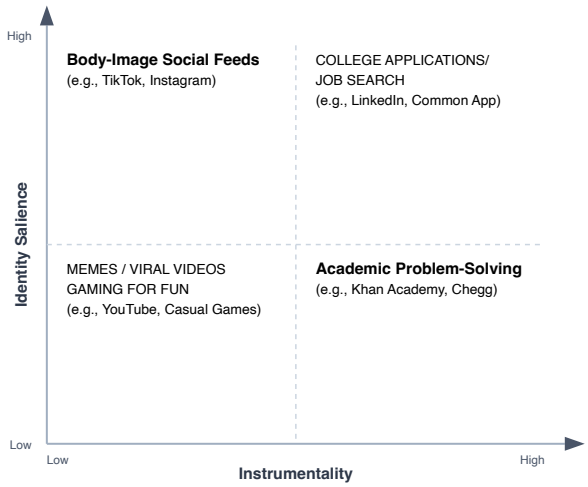

Figure 1: Applications of Calibrated Dependence framework.

by individual cognitive traits.

RQ3 (Mapping the Curve - Identity): How does this curve shift in the context of social media and identity? A parallel experiment in our FeedLab, which simulates a next-generation generative media environment, will measure the impact of AI on affective well-being and social comparison.

RQ4 (Intervention & Generalization): Can we design and validate interventions that nudge users toward the optimal "sweet spot" of calibrated dependence? By testing a theory-driven "Calibrated Dependence Coach" in both contexts, we will derive generalizable design principles for building more responsible and autonomy-preserving AI systems.

2 Background and Intellectual Innovations

This proposal makes notable intellectual contributions across multiple areas, advancing the state-of-the-art in human-computer interaction, the learning sciences, and computational social science.

2.1 The New Default: An Unscaffolded, AI-Saturated Reality

The digital environments of university students are becoming saturated with generative AI. This transformation is happening on two fronts: active production and passive consumption. On the production side, recent surveys confirm that a majority of U.S. university students now use generative AI in their coursework daily [1, 2]. Students leverage AI to offload cognitive effort, from summarizing readings and checking grammar to generating initial drafts [3]. On the consumption side, the social media feeds that shape their social lives and identities are increasingly populated with AI-generated and algorithmically-optimized content, from synthetic influencers to AI-powered beauty filters [5, 4]. Given the way things are currently, it is safe to assume that the dependence on AI in both production and consumption will only increase in the coming years.

Critically, our understanding of this new reality is poor. We lack robust, observational data on how students are truly engaging with these tools. Self-report surveys, while useful, are often unreliable; recent work reveals a significant "say-do" gap, where reported AI usage can be vastly different from actual, logged behavior [6]. This discrepancy may arise because students are unaware they are consuming AI content or are unwilling to report behavior they fear may be judged. This

highlights a critical need for observational methods to accurately map the true prevalence of both AI production and consumption. Our work addresses this gap by providing the first large-scale, longitudinal, and multi-modal dataset of university students' real-world interactions with generative AI across both academic and social contexts.

2.2 The Double-Edged Sword Of AI Assistance

The potential benefits of this new reality align with established principles in the Learning Sciences. When used effectively, AI can function as a powerful cognitive scaffold, offloading lower-order tasks to free up mental resources for higher-order thinking [7]. Research has shown that AI assistance can boost productivity in complex tasks like coding and professional writing [8]. This aligns with key principles from the Learning Sciences, where recent work demonstrates that intentionally designed AI systems can deliver structured, real-time feedback and explanations, acting as effective scaffolds that enhance, rather than supplant, students' cognitive processes [9, 10, 11].

In the social and identity domain, generative tools offer unprecedented avenues for creative self-expression. The ability to create personalized avatars and synthetic media allows users to safely explore different facets of their identity—experimenting with appearance, style, and social roles in a space free from physical-world consequences. This "identity play" can be particularly valuable for adolescents and young adults navigating who they want to become [12]. Furthermore, this technology can be empowering; embodying an avatar with idealized characteristics (e.g., being taller or more confident) can actually change the user's behavior to be more consistent with that trait, a phenomenon known as the Proteus Effect [13].

However, the "in-the-wild" use of generative AI often lacks this intentional design, leading to significant risks. The primary concern, echoed across recent studies, is the threat of over-dependence, which can erode the very skills education aims to build. Several studies recently have found evidence of the harms caused by over-dependence. Fan et al. [14] term this as "metacognitive laziness": students using ChatGPT produced well-polished essays but engaged less with source material and their own evaluative processes, ultimately learning less. Similarly, [15] find a significant decline in test performance due to over dependence on AI tools on a large sample of undergraduate students. This aligns with classic work on automation dependence, which warns that over-reliance on a system leads to skill atrophy and a loss of situational awareness [16]. The risk is that students offload not just tedious work but the "desirable difficulties" that are essential for deep learning and critical thinking. This can impact not only cognitive skills but also a student's sense of autonomy and academic identity, blurring the lines of authorship and ownership over their intellectual learning.

In the social sphere, the same mechanisms that enable positive identity exploration can become harmful. The constant exposure to algorithmically-perfected, AI-generated images of people (including idealized versions of oneself) can intensify social comparison processes, which are strongly linked to body dissatisfaction and negative mental health outcomes [17]. The very act of embodying a "perfected" avatar can widen the gap between one's digital and physical self, potentially exacerbating body image concerns. The risk is that students offload not just tedious work but also the critical cognitive and affective processes essential for their intellectual and personal development.

2.3 Bridging The Gap: From Automation Trust To AI-Supported Self-Regulation

To navigate this double-edged sword, we synthesize two major theoretical traditions. From HCI and human factors, the literature on trust calibration provides a crucial starting point. Researchers have long argued that the goal is not to maximize trust in automation, but to calibrate it appropriately to the system's true capabilities [18]. An effective user must learn when to trust an AI—whether

it's an academic hint or the authenticity of a social media image—and when to remain skeptical.

However, trust calibration alone is insufficient. In an educational context, the user is not just an operator but a learner, whose primary goal is to build their own internal capabilities. This is the domain of the Learning Sciences and the theory of Self-Regulated Learning (SRL). SRL describes the processes by which learners metacognitively, motivationally, and behaviorally manage their own learning [19, 20]. A successful student plans their approach, monitors their understanding, and reflects on their performance. The core challenge of generative AI is that its frictionless, high-quality output can short-circuit this entire regulatory loop.

Existing SRL models were not designed for a world where every learner has an instant, powerful, external cognitive partner. This project posits that we need a new model: AI-supported Self-Regulation. This model must account for a new set of self-regulatory skills: knowing when to offload a task to AI versus when to engage in productive struggle; critically evaluating and integrating AI output rather than passively accepting it; and managing the affective and identity-related consequences of using AI.

2.4 Intellectual Innovation: The Calibrated Dependence Framework

This proposal introduces the Calibrated Dependence (CD) framework as the first comprehensive, empirically testable model of AI-supported Self-Regulation. It bridges the gap between classic automation theory and SRL by providing a unified metric—the CD-Score—that jointly optimizes for performance (Utility Gain) and the preservation of learner agency (Autonomy Penalty). By modeling the relationship between AI assistance and the CD-Score as an inverted-U curve, we can move beyond the simplistic "ban AI vs. embrace AI" debate. Instead, we can empirically identify the "sweet spot" of optimal assistance for different tasks and learners. The intellectual innovations of this project are four fold:

1. A First-of-its-Kind Observational Dataset: This project will produce the first large-scale, longitudinal dataset of university students' real-world production and consumption of generative AI across academic and social contexts, moving beyond unreliable self-reports.
2. A Novel Theoretical Synthesis: We are the first to explicitly link the HCI concept of trust calibration with the Learning Sciences framework of SRL to address the unique challenges of generative AI.
3. A Validated Measurement Instrument: The CD-Score and its operationalization provide a new, validated method for quantifying the complex trade-offs of AI use, creating tools that future researchers and instructional designers can adopt.
4. An Evidence-Based Intervention: We design a "Calibrated Dependence Coach" which is not just another AI tool, but a set of theory-driven, autonomy-preserving scaffolds. By testing these interventions, we will provide the first evidence-based design patterns for building AI systems that foster, rather than hinder, self-regulated learning.

3 Methods

In this section, we will first describe our data collection methodology (Section 3.1), the toolkit we plan to build (Section 3.2), followed by our three methodological contributions: (i) detecting the prevalence of Generative AI content in various facets of production and consumption in students' lives (Section 3.3), (ii) defining and operationalizing calibrated dependence (Sections 3.4, 3.5), and, (iii) designing interventions to identify the sweet spot for calibrated dependence (Section 3.6).

3.1 Sampling And Data Collection

The project will enroll 1,000 U.S. undergraduates aged 18–22, a cohort that is both legally autonomous and at the forefront of intensive AI adoption for coursework and social media [21]. Recruitment will follow a dual-track strategy designed to maximize demographic breadth while preserving analytic comparability across campuses.

Panel track. Five hundred participants will be drawn from the higher-education sub-panels maintained by professional panel providers like YouGov and Ipsos Knowledge Panel. These probability-based panels recruit members through address-based sampling and routinely verify current college enrollment, allowing me to obtain a stratified cross-section of public- and private-institution students nationwide. Quotas will balance gender identity (50% women, 45% men, 5% non-binary/other), race/ethnicity (with deliberate oversamples of Black, Latino, and first-generation students), and broad field of study (STEM vs. humanities/social science), mirroring distributions reported in the 2024 National Center for Education Statistics digest [22].

Social media ads track. To reach students outside commercial panels and bolster representation from community colleges, we will recruit an additional 500 undergraduates via Instagram and Snapchat ads geo-targeted to a rotating list of 40 university ZIP codes. Ads invite users to a Qualtrics screener that verifies age, enrollment, and device compatibility, after which informed consent is obtained electronically. Prior work shows that social-ad sampling of this kind yields response pools comparable on key socio-demographic variables to traditional campus-flyer methods but at roughly one-third the cost [23].

The PI brings extensive experience in working with survey firms and social media ads to recruit survey participants [24, 25, 26, 27].

3.2 Study Tools And Measurement Instruments

This proposal advances two tightly coupled strands of work. First, we capture how university students produce and consume generative AI content in their study, work, and leisure (the instrumentation layer). Second, we test and refine the theory of calibrated dependence, the sweet-spot at which external intelligence maximizes benefit without eroding self-direction (the intervention layer). A single, privacy-preserving technical stack supports both strands, guaranteeing that observational insights inform the experimental designs and that the experiments, in turn, yield design guidelines students can immediately adopt.

3.2.1 The Instrumentation Layer: Capturing Real-World Use

To map prevalence and patterns of use, every participant installs a cross-browser extension and an Android accessibility service (iOS Screen Time hooks are requested where available). The extension records three classes of events: the domain, and timestamp of any visit to a recognized LLM (e.g., chat.openai.com, chatgpt.com, claude.ai, gemini.google.com); the user prompts and LLM responses, redacted locally for PII by a lightweight on-device filter; aggregate counts of clicks on LLM functions like "regenerate" or "upload file" functions that often signal deeper reliance. The accessibility service on our app mirrors this strategy for mobile social-media apps, logging the identifier of each piece of content viewed, the dwell time, and what in-app settings are toggled. Both tools allow us to export formatted JSON files to the device and upload them nightly; no raw screen content or keystrokes ever leave the device.

Participants can supplement passive logs with voluntary data donations through a secure web portal modeled after the Social Media data donation frameworks like PORT [28]. One tab walks students through exporting their TikTok, YouTube, or Instagram data archive [29]; another provides

a one-click export of their ChatGPT or Gemini usage history (or which ever LLMs are prominent at the time of study). The portal parses these archives client-side, strips personal identifiers, and stores only derived statistics such as frequency of use, average session length, prominent prompt topics, and a prompt-length histogram.

3.2.2 The Intervention Layer

To study calibrated dependence, our project relies on a browser-based research portal. The portal will have ability to run personalized surveys for various tests like the Positive and Negative Affect Schedule (PANAS) [30], the Collegiate Learning Assessment test [31], etc. The core feature of the portal is two web apps behind the same login:

LearnLab provides discipline-agnostic reasoning scenarios drawn from the Collegiate Learning Assessment (CLA+) library [31]. A back-end gateway will route hint requests either to (a) no service, (b) an open-source gradient-boosted "similar student" suggester, or (c) commercial LLM APIs (like OpenAI GPT-4o or Gemini 2.5), thereby switching assistance on or off with a single flag. Because all hint traffic is proxied through the server, the portal can simultaneously log every request and block tabbing to external LLM sites during task blocks. Every keystroke; which hints are requested; time spent editing; time between receiving a hint and submitting; final solution quality scored automatically by the CLA+ rubric.

FeedLab is designed as a sophisticated simulation of a next-generation social media environment where the lines between real, algorithmically curated, and user-directed generative content are blurred. The FeedLab environment mimics the UI of popular short-form video apps like Tiktok is populated by a dynamic mix of content: (i) Baseline Content: A large pool of real-world, non-AI videos drawn from public social media datasets [32, 33], covering typical genres like dance, comedy, and tutorials; (ii) Generic AI Content: A corpus of AI-generated videos and images featuring synthetic "supermodels" and influencers, created using state-of-the-art models. This simulates exposure to non-personalized, idealized AI creations; (iii) Personalized Generative Content: This is the core innovation of the FeedLab. Upon entering the study, each participant will create a personal, photorealistic digital avatar using a state-of-the-art character creation tool. This avatar, which the user can customize to resemble themselves, serves as a safe and ethical proxy for their own image. Within the FeedLab, participants can then use a built-in, prompt-driven AI tool to generate novel content featuring their avatar (e.g., "Show my avatar wearing this season's fashion," or "Create a video of my avatar performing this dance.").

The AI assistance within FeedLab includes a recommender system that ranks and surfaces all three types of content, as well as the generative tool that allows for active, on-demand content creation. This design allows us to experimentally manipulate not just what users see, but also what they can create, providing a rich environment to study the nuances of utility and autonomy.

Figure 2 shows the four-phase experimental protocol. The study begins with (1) participant onboarding and baseline measurements, followed by (2) a period of naturalistic observation of their digital behavior. Participants then engage in (3) a controlled experiment with multiple conditions, and conclude with (4) a follow-up session to assess knowledge retention and provide a full debriefing.

3.3 Identifying Generative AI Content

A central goal of this project is to understand not only how students use generative AI tools but also how they are passively exposed to AI-generated content within their information ecosystems. While our instrumentation can effectively log active use (e.g., visits to ChatGPT), identifying consumed content requires a robust, multi-modal classification pipeline. We build this based on data collected

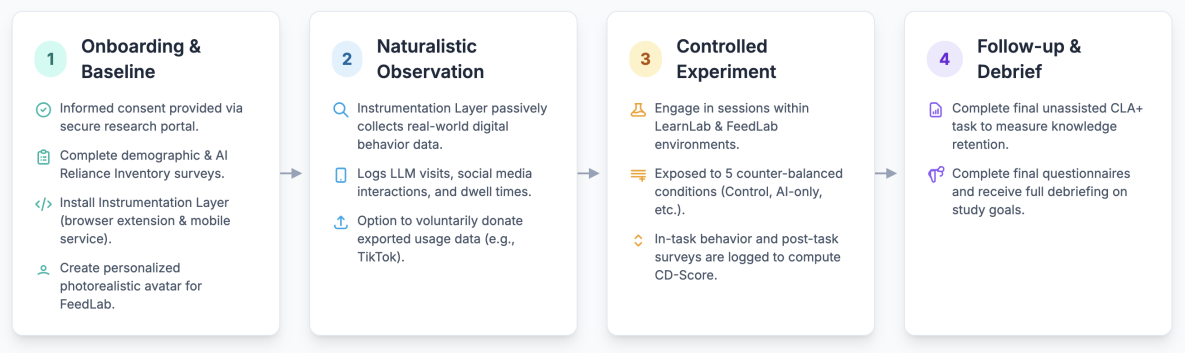

Figure 2: Four phases of our experimental protocol.

by the instrumentation layer (Section 3.2.1).

Our strategy does not reinvent the wheel; it builds on top of the open-source community already pushing the frontier of multimodal AI-content detection. We will leverage, fine-tune, and extend state-of-the-art public models, such as using text detectors inspired by GPTZero and DetectGPT [34], frequency-domain image detection [35], and 3-D CNNs for video deepfake cues [36] or other personalized AI video content, to assemble a single, extensible framework tuned to our project needs. Our aim is not a flawless, universal "AI detector" (an unattainable ideal) but a flexible tool that spots the statistical and stylistic fingerprints of the models students actually meet today (GPT-4/O3, Gemini 2.5, Llama 4, and their successors). Because different modalities dominate different risk contexts, the pipeline is explicitly multimodal. Crucially, we will continuously refresh the training set with content harvested from our own longitudinal panel, ensuring ecological validity and rapid adaptation to shifts in generative AI content generation. With a dedicated budget line item for manual annotation, compute, and synthetic-data generation, the system will provide lower-bound but objective estimates of students' exposure to generative content and reveal how those exposures ebb and flow over the life of the project.

3.4 Measuring Calibrated Dependence: Empirical Strategy

We define calibrated dependence (CD) as the dynamic equilibrium in which external intelligence maximizes functional benefit while preserving the user's experienced and demonstrated autonomy. The equilibrium is captured by a continuous CD-Score:

$$CD\ Score = Utility\ Gain - Autonomy\ Penalty \tag{1}$$

Utility gain is the difference between performance (or affect) achieved with assistance and the participant's own baseline under minimal help. *Autonomy penalty* is the composite cost measuring multiple variables such as perceived control, behavioral initiative, and cognitive reflection time. A higher CD-Score indicates better calibration; the score has an inverted U-shape curve and starts to decrease when help begins to harm sovereignty more than it helps performance. The empirical aim of this work is to locate the peak and test how its position shifts with task context and individual traits.

Figure 3 shows a visual model for understanding the crucial trade-off in designing effective AI assistance. The graph plots the relationship between the increasing 'Level of AI Assistance' (x-axis) and its overall impact (y-axis, a hypothetical score). We can see from the figure that simply maximizing AI help is a flawed strategy. The Utility line (green, dashed) shows that while the functional benefits of AI initially increase with more assistance, they are accompanied by a rising Autonomy Penalty (red, dashed), which represents the cost to a user's self-efficacy and knowledge

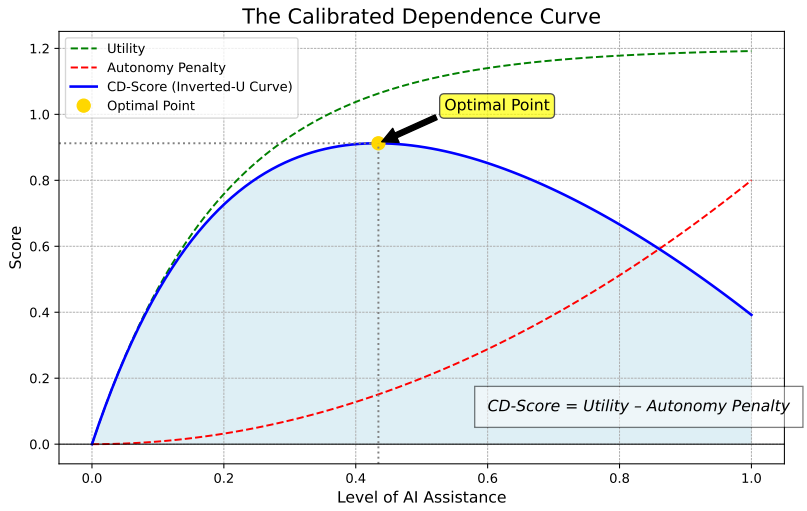

Figure 3: Sample calibrated dependence curve.

retention. The key insight is captured by the solid blue CD-Score curve, which represents the net effectiveness (Utility – Autonomy Penalty). Its inverted-U shape clearly demonstrates that after a certain point, too much assistance becomes detrimental, leading to a decline in overall performance. The peak of this curve highlights the design target: a 'sweet spot' of calibrated dependence where the user's functional gain is maximized, and their autonomy is preserved, preventing the pitfalls of over-dependence.

3.5 Operationalizing Calibrated Dependence

To operationalize the measurement of the CD-Score, we must translate the abstract concepts of Utility Gain and Autonomy Penalty into concrete, measurable variables within our two experimental contexts: student learning and social media consumption. Our methodology is designed to be multi-faceted, combining validated self-report scales with rich, unobtrusive behavioral data logged by our custom research platform (LearnLab and FeedLab). We present one concrete method for calculating the CD-Score, but the underlying toolkit is deliberately general. Researchers could implement alternative, equally valid operationalizations by swapping in different performance metrics, autonomy indicators, or task contexts without altering the core logic of calibrated dependence.

3.5.1 Measuring Calibrated Dependence In Student Learning

In the context of academic work, the tension between AI as a helpful tool and a cognitive crutch is particularly acute.

Utility Gain. The utility of AI assistance in an academic task is more than just getting the right answer. It also involves efficiency and, most critically, the durable learning that results. Therefore, our Utility Gain metric for the LearnLab context is a weighted composite of three distinct components:

1. Performance Improvement: To ensure a standardized, discipline-agnostic measure of performance, we will use scenarios from the Collegiate Learning Assessment+ (CLA+). The CLA+ is a nationally recognized instrument for assessing critical thinking and written communication skills in higher education [37]. Each participant will establish a baseline performance score on unassisted tasks. The Performance Improvement component is then the change in

their CLA+ score on an experimental task relative to their own baseline. This within-subject comparison normalizes for pre-existing differences in academic skill.

2. Efficiency Gain: AI assistance can significantly reduce the time required to complete a task. Using the event logs, we will measure the total time-on-task for each scenario. The Efficiency Gain is calculated as the proportional reduction in completion time compared to the participant's baseline, rewarding tools that make students faster without sacrificing quality.
3. Knowledge Retention: This is a crucial and novel component of our utility measure. A high score on a task where AI was heavily used may be misleading if the student has not actually learned the underlying concepts. To measure this, we will administer a surprise follow-up assessment one week after the experimental session, asking participants to solve a similar problem without any assistance. The score on this retention task provides a direct measure of durable learning.

These three components will be z-scored and combined into a single Utility Gain score. This composite approach provides a holistic view of "benefit," rewarding AI assistance that leads to better, faster work that also results in lasting knowledge. The use of CLA+ scores is one way to operationalize the utility gain measurement. Depending on the sampling frame (e.g. sampling students from the same class room), we could develop a way to measure this beyond just standarized tests (such as CLA+) into regular course work the students use it for.

Autonomy penalty. The Autonomy Penalty quantifies the cognitive and agentic cost of AI assistance. It is measured through a combination of self-perception and observed behavior.

1. Perceived Control: Immediately following each task, participants will complete the three-item autonomy subscale of the Basic Psychological Need Satisfaction scale [38]. This widely validated instrument from Self-Determination Theory provides a direct measure of the student's felt sense of agency and self-direction during the task.
2. Behavioral Initiative: We will use our detailed interaction logs to create a behavioral measure of initiative. This goes beyond simply counting hint requests. We will calculate an "Originality Ratio" by measuring the textual difference (e.g. Levenshtein distance or a semantic measure) between the AI's suggestions and the student's final submitted text. A low distance (i.e., near-verbatim copying) indicates low initiative and incurs a higher penalty. Similarly, the frequency of actions like ignoring a suggestion or manually editing a large portion of AI-generated text will contribute to a higher initiative score.
3. Cognitive Reflection: Over-reliance on AI is often characterized by a lack of critical reflection. We will use event timestamps to measure "Deliberation Time": the duration between when an AI hint is displayed and when the student takes their next action (e.g., typing, clicking). A consistently short deliberation time suggests superficial processing and "blind acceptance," thus incurring a higher autonomy penalty.

The three components of the Autonomy Penalty: Perceived Control (from a survey), Behavioral Initiative (from log data), and Cognitive Reflection (from time data) will be combined into a single composite score of converting them into z-scores and taking an average. As a crucial validation step, we will calculate the internal consistency of our three-part penalty measure using Cronbach's alpha. A high alpha score would indicate that all three components are reliably measuring the same underlying construct of "autonomy cost." If the reliability is high, the equal-weighting approach is justified. If one component proves to be less reliable, we may explore using reliability-based weighting in our final model.

3.5.2 Measuring Calibrated Dependence On Social Media

In a hyper-personalized context such as social media consumption, the concepts of Utility Gain and Autonomy Penalty become more complex, encompassing creative expression and the psychological impact of idealized synthetic selves.

Utility Gain. The "utility" of this advanced feed on FeedLab is measured as a composite of affective experience and creative empowerment.

1. Affective Shift: As a core measure of hedonic value, we will continue to use the Positive Affect subscale of the PANAS [30], administered before and after each session to capture the immediate emotional impact of the feed.
2. Creative Agency: A key benefit of generative AI is the power to create. We will quantify this by logging the frequency and complexity of a user's interactions with the avatar generation tool. A higher Creative Agency score will be assigned to users who frequently generate novel content, use more complex prompts, and spend more time customizing their creations. This metric captures the "upside" of generative tools, measuring their use as a medium for self-expression rather than just passive consumption.

Autonomy Penalty. Autonomy Penalty in this context must capture not only mindless consumption but also the specific psychological costs associated with exposure to idealized, AI-generated bodies and personalized avatars.

1. Perceived Control: The validated three-item autonomy subscale from Self-Determination Theory [39] will be used for measuring a user's felt sense of agency within the environment.
2. Mindful Consumption vs. Passive Entrainment: We will continue to use behavioral and gaze-based metrics to distinguish mindful interaction from passive scrolling. A high scroll velocity, and a low frequency of "pattern-breaking" behaviors (e.g., using search, visiting a profile) will all contribute to a higher penalty, indicating a state of algorithmic entrainment.
3. Social Comparison Intensity: Exposure to idealized images (especially personalized ones) is known to trigger social comparison. To measure this directly, we will administer a modified version of the Upward Social Comparison Scale [40] after each session. The scale will be adapted to specifically ask about comparisons to the content viewed in the feed, including the AI-generated models and the user's own modified avatar. A higher score on this scale represents a significant psychological cost and will be a major component of the Autonomy Penalty.

By operationalizing the CD-Score with these nuanced metrics, we can move beyond a simple analysis of "screen time" to a sophisticated evaluation of how hyper-personalized, generative media impacts student well-being. This approach allows us to test whether our Calibrated Dependence Coach interventions (next section) such as a prompt that asks, "How does seeing your avatar in this way make you feel?" can effectively mitigate the risks of social comparison while still allowing for the benefits of creative expression.

3.6 Intervention: The Calibrated Dependence Coach

The ultimate goal of this research is not merely to map the curve of calibrated dependence, but to develop and test interventions that help students locate and maintain the optimal balance between AI assistance and their own autonomy. Having operationalized the CD-Score, we can now define the experimental condition designed to actively improve it. Our intervention takes the form of a "Calibrated Dependence Coach," a set of theory-driven, just-in-time prompts and interface modifications delivered within our LearnLab and FeedLab environments. This "Coach" condition represents the highest level of assistance in our experimental design, testing the hypothesis

that targeted, metacognitive scaffolds can shift users up the inverted-U curve, increasing their net CD-Score.

Theoretical Grounding in Learning Sciences. The design of the Calibrated Dependence Coach is explicitly grounded in decades of research from the Learning Sciences on scaffolding and Self-Regulated Learning (SRL). Traditional SRL models focus on how learners monitor and control their own cognition and motivation [19]. The ubiquitous availability of generative AI introduces a new, external agent into this process, creating a novel need for what we term AI-supported Self-Regulation. Our Coach is designed to provide the minimal necessary scaffolds to help students regulate this new relationship effectively. It does not provide answers but instead prompts the user to engage in the metacognitive processes such as planning, monitoring, and reflection, that are often short-circuited by frictionless AI assistance. The interventions are designed to be "fading," providing support that can be internalized by the user over time, a core principle of effective scaffolding [41].

3.6.1 Operationalizing The Coach Interventions

The Coach will manifest as context-aware UI elements and prompts that are triggered by specific user behaviors logged by our system. The interventions are designed to be subtle, enhancing rather than disrupting the user's workflow. Table 1 showcases these interventions, detailing the specific prompts and nudges for both the LearnLab and FeedLab contexts. Each intervention is explicitly linked to the component of the Calibrated Dependence Score it is designed to influence, connecting our theoretical framework directly to the practical design of the Coach.

Table 1: Calibrated Dependence Coach Interventions

Context	**Intervention**	**Description & Example Prompt**	**Target Construct**
LearnLab (Academic)	Pre-Hint Reflection	Inserts a micro-pause before showing a hint. *"What is one possible next step you could take on your own?"*	Cognitive Reflection
	Post-Hint Justification	Asks user to explain their use of an AI suggestion. *"In one sentence, why does this improve your work?"*	Knowledge Retention
	Source Transparency	Adds a footnote to AI hints. *"Suggestion based on: a 2018 academic paper."*	Critical Evaluation
FeedLab (Social Media)	Creative Scaffolding	Suggests more complex prompts for avatar creation. *"Try adding a style, like 'in a vintage postcard'."*	Creative Agency
	Social Comparison Circuit-Breaker	Prompts reflection after viewing idealized AI content. *"This is AI-generated. How does it make you feel?"*	Social Comparison
	Reality Calibration Nudge	Periodically labels a non-AI video. *"This clip was not created or modified by AI."*	Mindful Consumption

3.6.2 Finding The Optimal Point

The "optimal point" or "sweet spot" on the Calibrated Dependence Curve (see Figure 3) is not a fixed value but a state we hypothesize our intervention can help users achieve. To systematically test our hypotheses, each participant will engage with the LearnLab and FeedLab environments under five distinct, counter-balanced experimental conditions. These conditions are designed to isolate the effects of algorithmic curation, generative AI assistance, and our coaching intervention:

1. Control: Participants experience the tasks without any AI or algorithmic support. In LearnLab, problems are presented in a fixed, predetermined order with no hint system. In FeedLab, videos are shown in a simple chronological order without ranking or generative tools. This establishes a pure baseline for performance and autonomy.
2. Algo-only: This condition introduces algorithmic curation without direct generative AI tools. In LearnLab, the platform uses a recommender algorithm to adaptively sequence problems based on the participant's performance. In FeedLab, the video feed is ranked by our recommender system (learned from user's own content).
3. AI-only: This condition introduces generative AI tools without algorithmic curation. In LearnLab, participants can request hints from the LLM, but problems are presented in a fixed order. In FeedLab, they can use the avatar and content generation tools, but the feed remains chronological. This isolates the effect of generative assistance.
4. AI+Algo: This condition represents the modern, fully-featured digital environment, combining both algorithmic curation and generative AI assistance. This allows us to measure the interaction effect between the two systems.
5. AI+Algo+Coach: This is the key intervention condition. It includes all the features of the AI+Algo condition, but with the addition of the Calibrated Dependence Coach and its associated nudges and prompts. This tests the efficacy of our designed scaffolds.

By analyzing the data from these five experimental conditions, we will model the full inverted-U curve. The "optimal point" will be empirically identified as the peak of this curve. Our primary hypothesis is that the average CD-Score in the AI+Algo+Coach condition will be significantly higher than in the AI+Algo condition, demonstrating that our theory-driven interventions successfully move users closer to this optimal point. By comparing the curve parameters across different user segments (e.g., based on their initial AI Reliance score), we can further understand how the optimal level of assistance and the effectiveness of the Coach may vary for different individuals.

3.7 Challenges

This project anticipates several key challenges and incorporates proactive strategies to address them. The primary technical hurdles involve data collection in a dynamic digital ecosystem. We recognize that the rapid evolution of generative models creates an "arms race" for AI detection. Our mitigation strategy is not to build a single, perfect detector, but rather a flexible, retrainable classification pipeline that leverages state-of-the-art open-source models. To ensure accuracy, we will continuously validate our automated classifiers against a ground-truth baseline established through the manual annotation of a random subset of the data.

Methodologically, we acknowledge two core challenges: the validity of our novel CD-Score and the ecological validity of our experiments. The validation of the CD-Score is a planned component of the research; we will use standard psychometric analyses to test the internal consistency of its composite parts. Should the components not cohere into a single metric, analyzing them as separate outcome variables is a planned alternative that will still yield valuable insights into the distinct ways AI impacts user perception, behavior, and cognition. We address the challenge of ecological

validity through the project's integrated two-part design. By directly comparing the behavioral patterns from our naturalistic observation phase with the results from our controlled experiments, we can empirically assess how well our findings generalize to real-world contexts.

Finally, we have a robust plan for the logistical challenges of participant recruitment and retention. To achieve our target sample size of 1,000 students, we will use a dual-track strategy combining established research panels with targeted social media advertising. To mitigate attrition over the course of the study, we have designed a carefully calibrated compensation structure with incremental payments and will over-recruit by 20% to ensure we maintain statistical power.

4 Broader Impacts

This project will deliver significant broader impacts by translating its research findings into tangible artifacts and foundational knowledge for the scientific community, and educators.

Focus on Students' Holistic AI Use. Current research on AI's impact often focuses narrowly on the active use of text-based tools like ChatGPT. This project makes a significant contribution by providing the first holistic view of students' AI interaction, capturing not only how they produce content for their academic work but also how they passively consume AI-generated visual media on social platforms. By developing methods to study these two distinct but interconnected fronts within a single, unified framework (Calibrated Dependence), this research will produce a foundational dataset that moves beyond unreliable self-reports. This will enable a more nuanced understanding of how AI is shaping the whole student—their intellect and their identity.

Practical Solutions for Calibrating Dependence. This project moves beyond critique to offer concrete, evidence-based solutions for mitigating the risks of over-dependence. The primary output is the Open Calibrated Dependence Toolkit, which includes an open-source SDK with validated design patterns. For students and educators, the toolkit provides practical strategies and self-assessment tools to analyze and manage their own AI usage, fostering a healthier AI ecosystem.

Interdisciplinary Contributions. This work stands at the intersection of several fields and is designed to make distinct contributions to each. For computational social science, it provides a novel, privacy-preserving methodology and a large dataset for studying human-AI interaction in the wild. For HCI, it introduces and validates a new theoretical construct (CD) and a novel metric (the CD-Score) for evaluating the UX of AI systems beyond simple task performance. For the learning sciences, it extends SRL to the new reality of powerful, external AI tools, creating a testable model for "AI-supported SRL" that accounts for both cognitive and affective regulation.

5 Results from Prior NSF Support

Garimella is a PI on a 'SaTC: CORE: Small: Towards a Privacy-Preserving Framework for Research on Private, Encrypted Social Networks' Award number 2318843. Period: 09/2023 to 09/2026. **Intellectual Merit:** The project develops privacy preserving techniques to collect data and study the prevalence of misinformation on encrypted social networks like WhatsApp. **Broader Impacts:** The project provides a first look at misinformation on a platform like WhatsApp and how it differs from other platforms like Facebook or Twitter. **Publications and products:** The project enabled research and tools to collect data from WhatsApp. WhatsApp Explorer, a tool to collect data at scale from WhatsApp, was partly funded by this project [24].

Garimella is a recipient of an NSF Convergence accelerator grant (Phases 1 and 2). The project was titled "Co-Insights: Fostering Community Collaboration to Combat Misinformation". Award

number: 49100421C0035, Period: 09/2021 to 09/2024. **Intellectual Merit:** The project provides a unique platform that enables community, fact-checking, and academic organizations to work together to respond effectively to misinformation targeting minority communities. **Broader Impacts:** The grant showed that misinformation claims often form around common themes, persistent stereotypes, and patterns of deception. **Publications and products:** Publications: [25]. Tools for data collection: WhatsViral [42] and Diaspora Watch [43].

6 Relevance to HCC

This proposal is submitted to the Human-Centered Computing (HCC) program and is directly aligned with its core mission to design and understand computing systems that amplify human capabilities. The project addresses a central challenge in modern HCI: how to design human-AI interaction to enhance students' cognitive and creative abilities while assessing and mitigating the risks of over-dependence. By integrating methods from the social and learning sciences with computer science, our work seeks to understand the complex interplay between students, generative AI systems, and the contexts in which they are used—from academic problem-solving to social media-based identity formation. The Calibrated Dependence framework provides a novel theoretical and methodological approach to this challenge, directly addressing HCC's interest in intelligent user interfaces, computing for creativity, and the social impacts of computing. By producing not just new knowledge but also evidence-based design patterns for interventions, this work embodies the HCC vision of research that both understands and actively shapes the future of HCI.

7 Education plan

The educational vision of this CAREER project is to translate the science of calibrated dependence into practical, evidence-based learning interventions. The plan has two synergistic components: (i) a critical AI literacy curriculum for the general university population, and (ii) a specialized design curriculum for students in technical disciplines. All development will be guided by collaboration with the university's Institute for Teaching, Innovation, & Inclusive Pedagogy.

7.1 Curriculum Development

All educational materials will be directly grounded in the project's primary research. The case studies, interactive modules, and design challenges will use anonymized, aggregated data and real-world scenarios uncovered in our study of student AI use. This ensures all content is authentic, relevant, and directly addresses the opportunities and challenges students face.

7.1.1 Critical AI Literacies For The University Community

This curriculum track is designed for the general student population and aims to foster the metacognitive skills necessary for calibrated dependence. The goal is not to prescribe when to use or avoid AI, but to equip students with a durable framework for making their own informed decisions. The pedagogical approach adapts the **Apt-AIR framework for epistemic cognition**, which focuses on learners' epistemic **A**ims, **I**deals, and **R**eliable processes [44]. The framework's focus on situational and metacognitive awareness is perfectly suited for navigating AI literacy as a knowledge domain for instructional development.

This framework will be delivered through two primary mechanisms: (i) "Calibrated Dependence Sparks": A series of 10-minute, interactive vignettes designed for easy integration into high-

enrollment first-year courses. Each "Spark" will present students with scenarios drawn from our research, such as a side-by-side comparison of the sources, credibility, quality and accuracy of outputs from a GPT-generated lab report and a student-edited version, or a prompt asking them to reflect on how AI-curated social media feeds engender emotional impacts. (ii) "AI & Autonomy" Module: A two-week intensive module, designed for courses like *Beyond the Hype: What AI can and can not do* (taught by PI Garimella), where students engage in a mini-self-study using the project's browser extension to visualize their own AI usage patterns (e.g., their own Utility-Penalty curves). A final reflection paper will ask them to analyze their habits and identify areas for recalibration, connecting personal experience to the theoretical framework.

7.1.2 Critical AI Engineering For Technical Students

For undergraduate and master's students in Information Technology & Informatics (IT&I) program and Master of Informatics program in the Library and Information Science department, this curriculum track focuses on translating the principles of calibrated dependence into technical practice. The objective is to train a new generation of engineers and designers who build systems that augment, rather than supplant, human agency. These modules will be integrated into existing courses such as Social Informatics, Interaction Design, and Human-Computer Interaction. Students will: (i) Learn to evaluate existing AI tools through the lens of calibrated dependence and the Apt-AIR framework[44, 45], identifying design patterns that encourage passive consumption versus active engagement while cultivating epistemic cognition and AI literacy. (ii) In project-based modules, students will use APIs from major generative models to build their own applications, explicitly support user autonomy. (iii) An annual campus hack-week will bring together interdisciplinary teams to build and test the kinds of interventions studied in our research.

7.2 Research Apprenticeships And Mentoring

Each academic year, we will recruit a cohort of six undergraduate and three master's students into a paid, credit-bearing research apprenticeship. To broaden participation in computing, at least half of the apprentices will be from groups historically under-represented in STEM. These students will help refine the data-collection tools, analyze telemetry data, and co-author short, public-facing ethnographic posts for the project website, providing direct training in both technical and science-communication skills. All mentoring will follow the structured *Entering Research* curriculum, which provides a national, evidence-based model for training undergraduate researchers, including modules on research ethics, effective communication, and career planning [46].

7.3 Open Dissemination And Community Building

To ensure the project's impact extends beyond our campus, all educational resources will be publicly disseminated. By Year 4, all curricular materials will be packaged and released under a Creative Commons license. The toolkit will include: (i) facilitator guides for the "Sparks" worked example activities and the "AI & Autonomy" module; (ii) anonymized, aggregated datasets from our research for classroom use; and (iii) turnkey React components for the "pause-and-reflect" interventions that instructors and developers can embed in their own applications. Starting in Year 4, we will run free, semi-annual webinars for educators at community colleges, K-12 schools, and university teaching centers. These sessions will guide instructors on how to adapt and deploy the toolkit for their own students. All training materials will be permanently archived and publicly accessible.

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Synergistic Activities

Throughout my career, I have consistently prioritized the integration and dissemination of knowledge alongside its creation, as demonstrated by my commitment to making research outputs accessible and useful to both academic and public sectors.

1. **Public Release of Research Tools and Code**: My projects often incorporate a strong public component, exemplified by the release of research tools and codes. For instance, under the NSF project 49100421C0035, my team developed a data donation tool for Facebook, which was documented in our publication [25]. We also launched a public dashboard available at kiran-research2.comminfo.rutgers.edu/plotly to facilitate broader access to our data. This platform enables users to replicate our research findings and explore additional insights, significantly enhancing the transparency and reproducibility of our work.

2. **Broader use of Data Donation Tools**: Another example of my research being of broader use is the open sourcing of our data donation methodologies for WhatsApp (funded by NSF projects 2318843 and 49100421C0035) [42, 24]. These tools have been adopted by researchers at institutions such as the University of Chicago and the University of Pennsylvania, aiding them in establishing their own data collection infrastructures and fostering a collaborative environment for scientific advancement.

3. **Collaborations Beyond Academia**: My work extends beyond traditional academic boundaries through active collaborations with fact-checkers, journalists, and NGOs. For example, a dashboard we created for monitoring WhatsApp data has been utilized by over 100 journalists and fact-checking organizations across India and Brazil [47]. This tool has not only facilitated more informed journalism but also contributed to more robust public discourse.

4. **Engagement with Media and Dissemination of Knowledge**: My research has regularly been featured in both national and international media outlets, such as The New York Times and The Washington Post [48, 49]. Through these engagements, I strive to promote the practical implications of our research findings to a global audience, thereby broadening the public understanding of technological impacts on society.

5. **Technology Deployment with Non-Profits and Companies**: In partnership with the non-profit tech company Meedan, we deployed a technology that matches claims sent via a tipline to previously verified fact-checks [50, 51]. Funded by NSF project 49100421C0035, this technology is now operational across 12 tiplines in six languages in India, supporting thousands of queries daily and significantly enhancing the efficiency and reach of factual verification efforts.

These activities not only illustrate my commitment to the broader impacts of my work but also highlight my efforts in fostering an ecosystem where academic research and practical application converge to serve a global community.